

Notes: Bordalo, Gennaioli & Shleifer (JPE, 2013)

Salience and Consumer Choice

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1 Big picture

- **Question.** How can one model context-dependent consumer choice when attention is drawn to attributes that stand out in the choice environment?
- **Core mechanism.** Consumers do not evaluate quality and price with fixed weights. They overweight the attribute that is more *salient*: the attribute whose value stands out more relative to the reference good in the choice context.
- **Choice context.** The reference good is usually the average product in the choice set. Later, the reference context also includes expected prices.
- **Key prediction.** Salience tilts choices toward goods with high quality-to-price ratios. A consumer focuses on the relative advantage of a good: either unusually high quality for the price or unusually low price for the quality.
- **Context dependence.** The same product pair can generate different choices when all prices shift, when a decoy is added, or when expected prices change.
- **Unified evidence.** The model organizes diverse phenomena: Savage’s car-radio problem, calculator-jacket savings, decoy and compromise effects, Thaler’s beer example, gasoline grade switching, and misleading sales.
- **Sales application.** A high regular price can serve as a decoy or anchor. A sale price may look like a bargain because the high full price changes the reference context and makes quality salient.
- **Economic message.** The paper turns “thinking in context” into a tractable attribute-weighting model that predicts when consumers trade up, trade down, or overreact to price changes.

2 Incremental contribution of the paper

- **From risky salience to riskless consumer choice.** Earlier salience models focused on lottery states and decision weights. This paper extends salience to deterministic goods with quality and price attributes.
- **A formal salience function.** The paper gives a parsimonious way to rank attribute salience using contrast and diminishing sensitivity. This allows the model to generate precise comparative statics rather than narrative psychology.
- **Quality/price ratio as a formal object.** The model derives, rather than assumes, the idea that consumers look for “good deals.” The quality-to-price ratio determines which attribute is salient and which product is favored.
- **Unification of context effects.** The same mechanism explains decoy effects, compromise-type choices, price-level effects, context-dependent willingness to pay, and sales promotions.
- **New empirical predictions.** The model predicts asymmetric decoy effects, differences between expected and unexpected price increases, and limits on misleading sales for standard goods sold by many sellers.
- **Distinct from loss aversion and rational inattention.** Unlike loss-aversion reference-point models, this framework can generate choices of extreme options and predicts directionally different responses to quality/price ratios. Unlike rational inattention models, attention is drawn ex post by salient contrasts.
- **Retail pricing implication.** The sales section contributes a behavioral theory of why retailers may display high regular prices and large discounts even when shoppers understand that sales occur.

3 Model primitives and data/measurement interpretation

3.1 Choice set, product attributes, and rational benchmark

A consumer chooses among goods

$$C = \{(q_k, p_k)\}_{k=1}^N,$$

where $q_k \geq 0$ is quality and $p_k \geq 0$ is price. Quality and price are both measured in dollars.

The rational benchmark is quasi-linear:

$$u_k = q_k - p_k. \tag{1}$$

Interpretation: In the rational benchmark, one dollar of quality improvement offsets one dollar of price. The salience model keeps this benchmark but allows attention to distort the relative weights on quality and price.

3.2 Reference good and salience ranking

The reference good has average attributes in the choice set:

$$\bar{q} = \frac{1}{N} \sum_{k=1}^N q_k, \quad \bar{p} = \frac{1}{N} \sum_{k=1}^N p_k.$$

For product k , quality salience is $\sigma(q_k, \bar{q})$ and price salience is $\sigma(p_k, \bar{p})$.

- Quality is salient if $\sigma(q_k, \bar{q}) > \sigma(p_k, \bar{p})$.
- Price is salient if $\sigma(q_k, \bar{q}) < \sigma(p_k, \bar{p})$.
- Attributes are equally salient if the two salience values are equal.

Interpretation: The reference good is not a consumed product. It is the contextual benchmark against which each product's quality and price are compared.

3.3 Properties of salience

The salience function satisfies two psychological properties.

1. **Ordering / contrast.** Salience increases when an attribute moves farther away from the reference value.
2. **Diminishing sensitivity.** A given absolute difference is less salient when the overall level of the attribute is higher.

A useful functional form is

$$\sigma(a_k, \bar{a}) = \frac{|a_k - \bar{a}|}{a_k + \bar{a}}, \quad a_k, \bar{a} \geq 0. \tag{4}$$

Interpretation: This form embodies Weber-style perception. A \$5 price difference matters more when prices are \$10 and \$15 than when prices are \$110 and \$115.

3.4 Salient utility weighting

A salient thinker overweights the salient attribute. Let $\delta \in (0, 1]$ measure the strength of salience distortions, with lower δ meaning stronger salience.

$$u_k^S = \begin{cases} \frac{2}{1+\delta}q_k - \frac{2\delta}{1+\delta}p_k, & \sigma(q_k, \bar{q}) > \sigma(p_k, \bar{p}), \\ \frac{2\delta}{1+\delta}q_k - \frac{2}{1+\delta}p_k, & \sigma(q_k, \bar{q}) < \sigma(p_k, \bar{p}), \\ q_k - p_k, & \sigma(q_k, \bar{q}) = \sigma(p_k, \bar{p}). \end{cases} \quad (5)$$

Interpretation: When quality is salient, willingness to pay for quality rises and price sensitivity falls. When price is salient, the consumer behaves as if price is especially painful and quality less important. The model is not about changing true preferences; it is about distorted attention weights.

3.5 Empirical measurement implications

- The model requires a choice context: which goods the consumer is comparing.
- It requires product attributes: price and some measure of quality or quality ranking.
- If exact quality levels are unavailable, the model still gives ordinal predictions using quality rankings and price movements.
- In settings with expected prices, expectations enter the reference context. This matters for sales, gasoline, and the beer example.

Interpretation: The paper is theoretical but empirically disciplined. Its empirical content lies in comparative statics: how demand changes when price levels, expected prices, or decoys change.

4 Models and theoretical specifications

4.1 Proposition 1: quality/price ratio determines salience

Under homogeneity of degree zero, a product's relative advantage is salient iff its quality-to-price ratio is above the reference good's quality-to-price ratio:

$$\frac{q_k}{p_k} > \frac{\bar{q}}{\bar{p}}.$$

Interpretation: This is the central bridge from psychology to economics. “Good deals” are not an extra utility term; they are products whose relative advantage draws attention.

4.2 Two-good wine example

Consider high-end wine $h = (q_h, p_h)$ and low-end wine $l = (q_l, p_l)$, with $q_h > q_l$ and $p_h > p_l$.

Quality is salient for both wines when

$$\frac{q_h}{p_h} > \frac{\bar{q}}{\bar{p}} > \frac{q_l}{p_l}. \quad (6)$$

If price is salient, the low-end wine is chosen when

$$\delta(q_l - q_h) - (p_l - p_h) > 0. \quad (7)$$

If quality is salient, the low-end wine is chosen when

$$(q_l - q_h) - \delta(p_l - p_h) > 0. \quad (8)$$

Interpretation: Price salience helps the cheaper good; quality salience helps the higher-quality good. Uniform price changes can therefore flip which wine looks like the better deal.

4.3 Proposition 2: goods along a rational indifference curve

If all goods lie on the same rational indifference curve, so that $q_k - p_k$ is constant across goods, the salient thinker chooses the good with the highest quality-to-price ratio:

$$k^* = \arg \max_{k=1,\dots,N} \frac{q_k}{p_k}.$$

Interpretation: When rational utility is held fixed, salience fully determines choice. The model predicts attraction to high value-per-dollar products.

4.4 Proposition 3: adding the outside option

When the outside option $(0,0)$ is included, salience can change because the reference good changes. The chosen good still tends to have an above-average quality/price ratio.

Interpretation: The outside option matters because not buying is itself a salient benchmark. It lowers average quality and price, making intermediate goods potentially look quality-salient.

4.5 Price-level effects and demand for quality

Because salience has diminishing sensitivity, uniform price increases can reduce the salience of price differences. At higher price levels, quality differences may become more salient.

Interpretation: This logic explains the wine-store versus restaurant example. The same absolute price difference is less salient in the restaurant, so quality becomes relatively more salient and the consumer trades up.

4.6 Decoy and compromise effects

A decoy is a dominated or unattractive good added to the choice set. In this model, a low quality/price decoy can shift the reference good and increase the salience of quality for a high-quality option.

Interpretation: The effect is asymmetric. A bad, expensive decoy can make a high-quality product more attractive, while a cheap low-quality decoy does not symmetrically boost the low-quality product in the same way.

4.7 Expected prices and expanded choice context

The model extends the choice context to include expected prices. If a consumer expects product k to have price p_k^e , the reference price can depend on both actual and expected prices.

Interpretation: This extension is crucial for field applications. Consumers enter stores, gas stations, and restaurants with expected prices. Surprise price changes are therefore more salient than expected price changes.

4.8 Proposition 5: expected versus unexpected price changes

The paper’s central comparative static is that expected and unexpected uniform price increases have different effects.

- If a uniform price increase is expected, it raises the reference price and reduces the salience of price differences. Consumers may become less price sensitive and trade up.
- If a uniform price increase is unexpected, the new price itself stands out relative to expectations. Consumers become more price sensitive and may trade down.

Interpretation: This is one of the model’s strongest empirical distinctions. The sign of the consumer’s quality response depends on whether the price increase was anticipated.

4.9 Proposition 6: context-dependent willingness to pay

For a good of quality q and expected price p^e , willingness to pay is

$$WTP(q|(q, p^e)) = \begin{cases} \delta q, & p^e \leq \delta q, \\ p^e, & \delta q < p^e \leq \frac{1}{\delta} q, \\ \frac{q}{\delta}, & \frac{1}{\delta} q < p^e \leq \frac{7}{2\delta} q, \\ \delta q, & p^e > \frac{7}{2\delta} q. \end{cases} \quad (14)$$

Interpretation: Moderate expected prices can raise willingness to pay because the consumer perceives quality as salient. Extremely high expected prices reverse the effect because price becomes salient again.

4.10 Propositions 7–8: misleading sales

For a nonstandard good, a retailer can display a high full price and a sale price. The high full price enters the reference context and can make the sale price look attractive.

- Proposition 7 shows that the retailer can charge up to the consumer’s maximum salient willingness to pay by choosing a sufficiently high full price.
- Proposition 8 shows that sales are more effective for high-quality goods than for low-quality goods.
- For standard goods sold by many sellers, misleading sales are less effective because price variation across sellers makes price salient.

Interpretation: This turns posted regular prices and discounts into attention-management devices. Sales can work not only by price discrimination but also by changing the reference context.

5 Identification and theoretical strategy

5.1 Type of identification

This is not an empirical causal-identification paper. Its identification is theoretical and behavioral: it identifies a mechanism that can rationalize multiple anomalies with a common set of assumptions.

Interpretation: The paper’s strategy is to show that a parsimonious salience model produces sharp, falsifiable comparative statics that differ from standard utility, loss aversion, rational inattention, and pure transaction-utility accounts.

5.2 Key assumptions that drive the mechanism

1. **Attributes are separable.** Quality and price enter utility additively.
2. **Context generates a reference good.** The reference good is initially the average product in the choice set.
3. **Salience increases with contrast.** Attributes that stand out receive attention.
4. **Salience has diminishing sensitivity.** The same absolute gap is less salient at higher levels.
5. **Salient attributes receive higher decision weight.** The consumer's valuation overweights the salient attribute.
6. **Expected prices can enter context.** In field settings, normal prices or expected prices affect what stands out.

5.3 How the paper separates salience from alternatives

- **Versus rational choice:** rational utility cannot generate preference reversals from irrelevant alternatives when true product payoffs are fixed.
- **Versus loss aversion:** loss-aversion models often predict compromise or middle options, while salience can produce attraction to extremes when they have salient advantages.
- **Versus transaction utility:** the model does not assume a direct taste for bargains; bargains matter because they shift attention.
- **Versus rational inattention:** attention is not chosen ex ante to reduce cognitive costs; it is drawn ex post by salient contrasts.

5.4 Falsifiable predictions

- Expected uniform price increases can increase demand for quality because price differences become less salient.
- Unexpected uniform price increases can decrease demand for quality because prices themselves become salient.
- Decoys should asymmetrically help high-quality goods when they reduce the reference good's quality/price ratio.
- Willingness to pay should vary with expected price context, as in the beer resort versus grocery-store example.
- Sales should be most misleading for nonstandard goods and less effective for standard goods sold by many sellers.

5.5 Threats and limits

- **Context measurement.** Researchers may not observe the consumer's true consideration set or expected prices.
- **Quality measurement.** Quality is often latent and may need to be inferred from prices, rankings, or product characteristics.
- **Parameter stability.** The distortion parameter δ may vary by consumer, product category, or attention environment.
- **Rank-based discontinuity.** The baseline model uses salience rankings, which can create discontinuities; the authors note that continuous salience weights preserve the main ideas.

- **Multiple attributes.** Real goods have many quality dimensions. The online appendix extends the model, but empirical applications must decide which dimensions enter attention.

6 Figure

6.1 Figure 1: Willingness to pay for q as a function of expected price p^e

Read:

- The horizontal axis is the expected price p^e .
- The vertical axis is willingness to pay for a good of quality q .
- When p^e is very low, the consumer's WTP is δq : price or low-value context dominates, so quality is not strongly valued.
- For moderate expected prices, WTP rises with p^e . A higher expected price makes quality appear more salient.
- Over an intermediate interval, WTP reaches q/δ , the high salient valuation of quality.
- When p^e is too high, WTP drops back to δq because price becomes salient again.

Interpretation: The figure captures the central nonmonotonicity of context-dependent willingness to pay. A higher expected price can make quality salient and raise WTP, but an excessively high expected price makes price salient and lowers WTP. This explains why a beer from a resort can command a higher reservation price than the same beer from a grocery store, but not if the resort price is perceived as outrageous.

Economic message: The reference context can act like an anchor, but only over a range. Moderate price expectations make consumers tolerant of higher prices; extreme expectations make price pain salient.

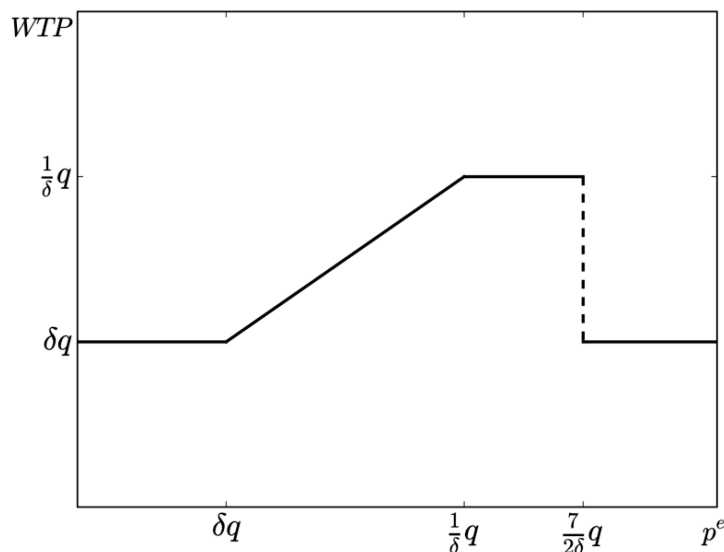


FIG. 1.—Willingness to pay for q as a function of expected price p^e

Figure 1: Willingness to pay for q as a function of expected price p^e .

7 Interpretive synthesis

7.1 What the model adds to consumer theory

The model says that consumer demand is shaped not only by product attributes but also by which attributes are made salient by the choice context. This is a middle ground between rational choice and unstructured behavioral anomalies: choices are distorted, but the distortion follows a stable rule.

7.2 Why quality/price ratios matter

Quality/price ratios summarize relative advantage. A high-end good can be attractive when its quality advantage is more salient than its price disadvantage. A low-end good can be attractive when its price advantage is more salient than its quality disadvantage.

7.3 Why expected prices matter

Expected prices define what is normal. A price increase that is fully expected may change the background level and reduce the salience of price differences. A surprise price increase sticks out and raises price sensitivity.

7.4 Why sales may mislead

A sale can be misleading when the high regular price affects the reference context. The discount then looks like a bargain, and the consumer overweights quality or undervalues price. The theory predicts that this works better for nonstandard goods than for standard goods with many sellers.

8 Short summary

- The paper models context-dependent consumer choice through salience.
- Goods have quality and price attributes; rational utility is $q - p$.
- Salience is measured by how much an attribute stands out relative to a reference good.
- The reference good is usually the average good in the choice set, and later also incorporates expected prices.
- A salient thinker overweights the more salient attribute.
- Homogeneous salience implies that quality/price ratios govern which attribute stands out.
- The model explains why consumers are attracted to goods that look like good deals.
- Uniform expected price increases can make quality more salient and induce trading up.
- Uniform unexpected price increases can make price salient and induce trading down.
- Decoys can shift the reference good and disproportionately increase demand for high-quality goods.
- Context-dependent willingness to pay explains Thaler's beer example.
- Misleading sales work by using a high regular price as a decoy or anchor.
- Sales are predicted to be less misleading for standard goods sold by many competing sellers.
- The paper's contribution is to unify many context effects with one tractable attribute-weighting mechanism.

9 Conclusion

9.1 Core conclusion

Bordalo, Gennaioli, and Shleifer develop a theory in which consumers evaluate goods by overweighting salient attributes. The model explains why choices depend on the comparison set, expected prices, and decoys, even when true quality and price are known.

9.2 Economic intuition

The intuition is that attention is drawn to relative contrast. If quality stands out more than price, consumers overvalue the high-quality option. If price stands out more than quality, consumers become price sensitive and favor the cheaper option. Because salience has diminishing sensitivity, a given dollar price difference matters less at high price levels.

9.3 Incremental contribution

The paper’s incremental contribution is to formalize context dependence in a portable way. Rather than treating each anomaly as separate, it shows how one salience mechanism can generate observed patterns in decoy effects, willingness to pay, gasoline substitution, and sales pricing. The key innovation is connecting salience to quality/price ratios and expected prices.

9.4 What the paper does not claim

The paper does not claim that consumers always choose irrationally or that salience replaces all other behavioral mechanisms. It also does not estimate δ structurally in a large dataset. Instead, it provides a theoretically disciplined framework that can be taken to data in settings where choice sets, product qualities, and expected prices are observable.

9.5 Final takeaway

The model’s final message is simple: consumers do not just ask whether a good is worth its price; they ask which attribute stands out in the current context. Firms, retailers, and policy makers can therefore shape demand by changing the comparison set, reference prices, and presentation of discounts.